

Two countries model

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Introduction

- Several central banks have large DSGE model featuring multiple countries/sectors and modelling the economic behavior of more than the simple representative agent, *e.g.* firms, intermediate producer, capital owners, banks, etc.
- The purpose of this chapter is to build a canonical dynamic, general equilibrium model of a two countries economy and to explain the observed international fluctuations by modifying the traditional modelling of the labor market in the two-country real business cycles model.
- This chapter then proposes a **medium scale** DSGE model with more rigidities and frictions than the previous one in order to replicate the observed economic dynamics.

Introduction

What are the difference between a small open economy and a two countries model?

- The small domestic open economy is derived as a limit case of the two-country model as it assumes exogenous processes for the foreign variables.
- In the two-country model the foreign variables are endogenous
- The terms of trade enters directly into the model
- Home bias in households preferences allows for real exchange rate fluctuations.
- Then, stronger propagation mechanisms which means that the model is not entirely explained by shocks that are exogenous.

Introduction

Why the labor market? (why not).

- The intuition is that labor-market search can be useful to understand the propagation of international fluctuations.
- Changes in the expected returns to search induce responses in search and recruiting activities, the effects of which are propagated through time via changes in the stock of employment.
- Given that the technology shock spills over to the other country, domestic and foreign firms start searching at the same time in anticipation.
- Employment then increases simultaneously and displays a hump-shaped profile in the two countries. This partially curtails the capital outflow from the country which does not benefit from the shock.

Objectives

- 1 Building a two countries DSGE model
- 2 Simulating the model. Confront the moments implied by the model to those from the data and simulate impulse response functions.

Jean Olivier Hairault. “Search Labor-Market in a Two-Good Two-Country Model”, Review of Economic Dynamics, 2002, 5, p.535-558.

Motivation

- International real business cycles (RBC) models have been extensively used to explain the observed aggregate correlations across countries.
- International fluctuations of consumption and output have received the main attention since the seminal work of Backus et al. (1992).
- Faced with the inability of the canonical one-good two-country model to replicate empirical features (Backus et al., 1992), according to which cross-country correlations of output are higher than the one of consumption (the so-called consumption correlation puzzle), numerous extensions have been proposed in the literature.

1. Motivation

- The aim of this paper is to explain the observed international fluctuations simply by modifying the traditional modelling of the labor market in the two-country RBC model.
- The intuition is that labor-market search can be useful to understand the propagation of international fluctuations arising from productivity shocks.

Pissarides, C. (1990). “Equilibrium Unemployment Theory”, Oxford: Basil Blackwell.

Merz, M. (1995). “Search in the Labor Market and Real Business Cycle,” *Journal of Monetary Economics* 36, 269-300.

Andolfatto, D. (1996). “Business Cycle and the Labor Market Search,” *American Economic Review* 86, 112-132.

1. Motivation

Build a model that is consistent with

- The persistence in unemployment and the asymmetric dynamic correlation between hours and labor productivity.
- output autocorrelation.
- international co-movements.

Plan

- 2 The model
- 3 Calibration
- 4 Simulation
- 5 Limits

2. The model

Assumptions

- Each country specializes in the production of a single good affected by technology shocks A .
- Households consume a CES basket of the two goods indexed by 1 (domestic good) and 2 (foreign good).
- They participate in the trade on the labor market.
- Employees work h units of time at a wage rate of w and unemployed workers devote e units of their time to seeking employment.

2. The model

A. The labor market search

The labor market is represented as an extension of the search and matching model as in Pissarides (1990).

- Let $N_{i,t}$ denote the number of employed workers at the beginning of the period t in the country i .
- As the population size is normalized to 1, $U_{i,t} = 1 - N_{i,t}$ is the measure of the unemployment rate
- Let $V_{i,t}$ be the total number of new jobs made available by firms during the period t , each vacancy incurring a flow cost equal to ω .
- The rate $H_{i,t}$ at which new job matches are formed is governed by an aggregate matching function:

$$H_{i,t} = \chi V_{i,t}^{\phi} (e_i U_{i,t})^{1-\phi} \quad i = 1, 2$$

χ is a measure of the efficiency of the matching process. e_i is the exogenous search effort per worker seeking employment, and $e_i U_i$ is the aggregate search effort.

2. The model

A. The labor market search

- Employment then evolves according to the law of motion:

$$\underbrace{N_{i,t+1}}_{\text{New employment}} = \underbrace{N_{i,t} - sN_{i,t}}_{\text{Remaining employment}} + \underbrace{H_{i,t}}_{\text{Hirings}}$$

Past employment Separation

where s is the exogenous rate of separation. Hence a job vacancy can at best become productive only after one period of time has elapsed.

- We define the vacancy filling rate as $m_{i,t} = H_{i,t}/V_{i,t}$ and the job finding rate (per unit of search) as $t_{i,t} = H_{i,t}/(e_i U_{i,t})$

2. The model

B. Households

- Each household participates in the labor market and consumes a CES basket C_i^c of the two goods indexed by 1 (domestic good) and 2 (foreign good),

$$C_i^c = \left[\gamma^{1/\theta} C_i^{(\theta-1)/\theta} + (1-\gamma)^{1/\theta} C_{j \neq i}^{(\theta-1)/\theta} \right]^{\frac{\theta}{1-\theta}} \quad i = 1, 2$$

where θ is the elasticity of substitution between the two goods and γ a parameters determining the share of the consumption basket and controlling for the ratio of imports to GDP.

- The price index in each country is:

$$P_i^c = \left[\gamma P_i^{1-\theta} + (1-\gamma) P_{j \neq i}^{1-\theta} \right]^{\frac{1}{1-\theta}} \quad i = 1, 2$$

where P_i is the price of the good i .

2. The model

B. Households

- We normalize the price of the domestic good (good 1) to 1 and denote by P the (relative) price of the foreign good (good 2).
- This latter corresponds to the terms of trade for the foreign country: the relative price of exported goods to imported goods. An increase of P will correspond to an appreciation (depreciation) of the terms of trade for the foreign country (domestic country).

2. The model

B. Households

- We assume perfect international risk sharing: households have access to contingent claims B at prices v , providing one unit of good 1 if the state A occurs (we denote by $f(A)$ the density function associated to the stochastic variable A).
- The country 1 representative household is assumed to maximize the expected discounted sum of its utility flows,

$$W_1^H(B_{t,1}) = \max \left\{ N_{1,t} U_{1,t}^n + (1 - N_{1,t}) U_{1,t}^u + \beta \int W_1^H(B_{1,t+1}) f(A) dA \right\}$$

subject to:

$$N_{1,t} P_{1,c}^c C_{1,t}^{c,n} + (1 - N_{1,t}) P_{1,t}^c C_{1,t}^{c,u} + \int v_t B_{1,t+1} dA \leq B_{1,t} + N_{1,t} w_{1,t} h_{1,t} \quad (\lambda_{1,t})$$

$$N_{1,t+1} = (1 - s) N_{1,t} + t_{1,t} e_1 (1 - N_{1,t}) \quad (\zeta_{1,t})$$

2. The model

B. Households

- The country 2 is analogue:

$$W_2^H(B_{2,t}) = \max \left\{ N_{2,t} U_{2,t}^n + (1 - N_{2,t}) U_{2,t}^u + \beta \int W_2^H(B_{2,t+1}) f(A) dA \right\}$$

subject to:

$$N_{2,t} P_{2,c}^c C_{2,t}^{c,n} + (1 - N_{2,t}) P_{2,c}^c C_{2,t}^{c,u} + \int v_t B_{2,t+1} dA \leq B_{2,t} + P_t N_{2,t} w_{2,t} h_{2,t} \quad (\lambda_{2,t})$$

$$N_{2,t+1} = (1 - s) N_{2,t} + t_{2,t} e_2 (1 - N_{2,t}) \quad (\zeta_{2,t})$$

- with $t_{i,t} = H_{i,t} / (e_i U_{i,t})$. We note that the wage rate w_i is defined in terms of the good produced at home.
- $C_i^{c,s}$, $s = n, u$ denotes the consumption of employed worker ($s=n$) and unemployed worker ($s = u$) in country i .
- $\lambda_{i,t}$ and $\zeta_{i,t}$, $i = 1, 2$ are the Lagrange multiplier associated to the budget constraint and the law motion for employment.

2. The model

B. Households

- the preferences are given by the following instantaneous utility function

$$U_i^s = \log C_i^{c,s} + \Gamma_s(h_i^s), \quad s = n, u, \quad i = 1, 2$$

with

$$\Gamma_n(h_i) = \sigma_n \frac{(1 - h_i)^{1-\eta}}{1 - \eta}$$

$$\Gamma_u(e_i) = \sigma_e \frac{(1 - e_i)^{1-\eta}}{1 - \eta}$$

- h_i denotes hours per worker and e_i denotes the time devoted to search.
- The parameter σ_s can be interpreted as reflecting differences in the efficiency of household's home production technology across different states of employment opportunities.

2. The model

B. Households

- Under perfect insurance market, one gets

$$\begin{aligned}C_{i,t}^{c,n} &= C_{i,t}^{c,u} \\ U_i^u - U_i^n &= \Gamma_u(e_i) - \Gamma_n(h_{i,t})\end{aligned}$$

The optimal conditions are

$$\begin{aligned}\frac{1}{C_{i,t}} &= P_{i,t}^c \lambda_{i,t} \\ v_t &= \beta \frac{\lambda_{i,t+1}}{\lambda_{i,t}} f(A)\end{aligned}$$

2. The model

C. Firms

- The two countries produce imperfectly substitutable goods with capital and labor: each country specializes in the production of a single good. The goods are produced with constant return to scale technology,

$$Y_i = A_i(K_i^c)^{1-\alpha}(h_i N_i)^\alpha, \quad i = 1, 2$$

- Investment to each country is an index of the two goods with the same structure as the consumption one. The capital index K^c evolves according to the following dynamic equation:

$$K_{i,t+1}^c = K_{i,t}^c(1 - \delta) + I_{i,t}^c$$

We assume there are capital adjustment costs $\varphi_{i,t}$:

$$\varphi_{i,t} = \frac{\Phi}{2} \frac{(K_{i,t+1}^c - K_{i,t}^c)^2}{K_{i,t}^c}$$

2. The model

C. Firms

- Fluctuations arise from persistent shocks to aggregate productivity,

$$\begin{pmatrix} \log(A_{1,t}) \\ \log(A_{2,t}) \end{pmatrix} = \begin{pmatrix} \rho & \rho^* \\ \rho^* & \rho \end{pmatrix} \begin{pmatrix} \log(A_{1,t-1}) \\ \log(A_{2,t-1}) \end{pmatrix} + \begin{pmatrix} (1-\rho) & -\rho^* \\ -\rho^* & (1-\rho) \end{pmatrix} \\ \times \begin{pmatrix} \log(A_1) \\ \log(A_2) \end{pmatrix} + \begin{pmatrix} 1 & \psi \\ \psi & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix}$$

- A_1 and A_2 are the means of the processes followed by $A_{1,t}$ and $A_{2,t}$, respectively.
- $\varepsilon_{1,t}$ and $\varepsilon_{2,t}$, which represent the innovations to productivity variables, share the same variance σ_A^2 and are such that $E(\varepsilon_{1,t}) = E(\varepsilon_{2,t}) = 0$, $E(\varepsilon_{1,t}\varepsilon'_{2,t}) = 0$.
- ψ is a positive parameter which determines the contemporaneous correlation between domestic and foreign technology shocks.

2. The model

C. Firms

- The program of each firm in country 1 is dynamic and consists of maximizing the expected discounted sum of profit flows,

$$W_1^F(K_{1,t}^c, N_{1,t}) = \max \left\{ Y_{1,t} - \varphi_{1,t} - P_{1,t}^c I_{1,t}^c - \omega_1 V_{1,t} - w_{1,t} h_{1,t} N_{1,t} + \int v_t W_1^F(K_{1,t+1}^c, N_{1,t+1}) dA \right\}$$

- subject to the following constraints:

$$\begin{aligned} K_{1,t+1}^c &= K_{1,t}^c(1 - \delta) + I_{1,t}^c \quad (\mathbf{q}_{1,t}) \\ N_{1,t+1} &= N_{1,t}(1 - s) + m_{1,t} V_{1,t} \quad (\boldsymbol{\xi}_{1,t}) \end{aligned}$$

2. The model

C. Firms

- The optimality conditions with respect to $\{I_{1,t}^c, V_{1,t}, N_{1,t+1}, K_{1,t+1}^c\}$

$$\tilde{\zeta}_{1,t} m_{1,t} = \omega_1$$

$$\tilde{\zeta}_{1,t} = \beta E_t \left\{ \frac{\lambda_{1,t+1}}{\lambda_{1,t}} \left(\frac{\partial Y_{1,t+1}}{\partial N_{1,t+1}} - w_{1,t+1} h_{1,t+1} + \tilde{\zeta}_{1,t+1} (1-s) \right) \right\}$$

$$q_{1,t} = P_{1,t}^c + \frac{\partial \varphi_{1,t}}{\partial I_{1,t}^c}$$

$$q_{1,t} = \beta E_t \left\{ \frac{\lambda_{1,t+1}}{\lambda_{1,t}} \left(\frac{\partial Y_{1,t+1}}{\partial K_{1,t+1}^c} - \delta P_{1,t+1}^c + q_{1,t+1} \right) \right\}$$

- The labor decisions (the first two FOC) depart from the ones gotten in the traditional two-country two-good RBC model. They are similar to the capital ones (the last two FOC). Because finding new workers takes time and effort, firms view their existing workforce as a capital asset.

2. The model

C. Firms

- The program of each firm in country 2 is dynamic and consists of maximizing the expected discounted sum of profit flows,

$$W_2^F(K_{2,t}^c, N_{2,t}) = \max \left\{ P_t[Y_{2,t} - \varphi_{2,t} - \omega_1 V_{1,t} - w_{2,t} h_{2,t} N_{2,t}] - P_{2,t}^c I_{2,t}^c + \int v_t W_2^F(K_{2,t+1}^c, N_{2,t+1}) dA \right\}$$

- subject to the following constraints:

$$\begin{aligned} K_{2,t+1}^c &= K_{2,t}^c(1 - \delta) + I_{2,t}^c \quad (q_{2,t}) \\ N_{2,t+1} &= N_{2,t}(1 - s) + m_{2,t} V_{2,t} \quad (\xi_{2,t}) \end{aligned}$$

2. The model

C. Firms

- The optimality conditions with respect to $\{I_{2,t}^c, V_{2,t}, N_{2,t+1}, K_{2,t+1}^c\}$

$$\xi_{2,t} m_{2,t} = P_t \omega_1$$

$$\xi_{2,t} = \beta E_t \left\{ \frac{\lambda_{2,t+1}}{\lambda_{2,t}} \left(P_{t+1} \frac{\partial Y_{2,t+1}}{\partial N_{2,t+1}} - P_{t+1} w_{2,t+1} h_{2,t+1} + \xi_{2,t+1} (1-s) \right) \right\}$$

$$q_{2,t} = P_{2,t}^c + P_t \frac{\partial \varphi_{2,t}}{\partial I_{2,t}^c}$$

$$q_{2,t} = \beta E_t \left\{ \frac{\lambda_{2,t+1}}{\lambda_{2,t}} \left(P_{t+1} \frac{\partial Y_{2,t+1}}{\partial K_{2,t+1}^c} - \delta P_{2,t+1}^c + q_{2,t+1} \right) \right\}$$

2. The model

D. Nash bargaining

- Wages and individual hours are bargained. The surplus generated by a match is shared between firms and households according to their bargaining power.
- Following Pissarides (2000), the bargaining process is given by the solution to the Nash criterion,

$$\max_{w,h} (\lambda \Omega^F)^\epsilon (\Omega^H)^{1-\epsilon}$$

- with $\Omega^F = \partial W^F / \partial N$ and $\Omega^H = \partial W^H / \partial N$. ϵ denotes the firm's share of a job's value. We deduce from this program the bargained wage and hours contract,

$$w_{1,t} h_{1,t} = (1 - \epsilon) \left(\frac{\partial Y_{1,t}}{\partial N_{1,t}} + \mu_{1,t} \omega_1 \right) + \epsilon \left(\frac{\Gamma_u(e_1) - \Gamma_n(h_{1,t})}{\lambda_{1,t}} \right)$$
$$\alpha \frac{Y_{1,t}}{N_{1,t} h_{1,t}} (1 - h_{1,t})^\eta = \frac{\sigma_n}{\lambda_{1,t}}$$

2. The model

D. Nash bargaining

- For firms in country 2 we have,

$$w_{2,t}h_{2,t} = (1 - \epsilon) \left(\frac{\partial Y_{2,t}}{\partial N_{2,t}} + \mu_{2,t}\omega_2 \right) + \epsilon \left(\frac{\Gamma_u(e_2) - \Gamma_n(h_{2,t})}{P_t\lambda_{2,t}} \right)$$
$$\alpha \frac{Y_{2,t}}{N_{2,t}h_{2,t}} (1 - h_{2,t})^\eta = \frac{\sigma_n}{P_t\lambda_{2,t}}$$

- with $\mu_i = V_i / (eU_i)$. The worker's wage bill is a weighted average of its contribution to output and its outside opportunities.
- The firm bargaining power ϵ determines the spread between the labor productivity and the wage bill.
- A decrease of λ_i , increases outside opportunities, giving an incentive to employees for working less and claiming higher wages.

2. The model

D. Nash bargaining

- Induced by a positive productivity shock in country 1, for instance, this effect tends to increase the product wage in country 2.
- On the contrary, this shock improves the terms of trade in country 2, leading to a decrease in its product wage.
- This provides country 2 firms a larger share of the match surplus, which not only increases their incentive for posting more vacancies, but may also curtail the capital outflow from country 2.

2. The model

E. Equilibrium

- Combining the FOC from households program and the budget constraint we have:

$$B_{i,t} + N_{i,t}w_{i,t}h_{i,t} = P_{i,t}^c C_{i,t}^c + \int v_t B_{i,t+1} dA$$

- The firm value function writes:

$$W_i^F = P_{i,t}$$

3. Calibration

A. Preliminary

- Share parameters for preferences and production are chosen such that means of ratios of aggregate times series are equal to analogous ratios for the theoretical economy's steady state.
- Several parameters are calibrated according to Backus et al. (1994) and in Andolfatto (1996).
- Consider a symmetrical calibration between the two countries.

3. Calibration

A. Shock processes and production technology

Shock processes

- ρ^* is fixed at 0.088 and ρ^* to 0.906 (Backus et al. (1994).)
- ψ is calibrated in order to get a correlation between technology innovations of 0.258
- $\sigma_A = 0.00852$ is set to match the volatility of output.

Production technology and costs

- The depreciation rate δ is set at 0.025.
- α is set in order to get a labor share of 64%.
- Φ , the capital adjustment cost parameter, is calibrated in order to replicate the volatility of investment.

3. Calibration

B. Preferences

- Following Backus et al. (1994), θ , the elasticity of substitution between domestic and foreign goods, is set at 1.5
- The discount factor β is set at 0.99. (average real interest rate of 0.04% annually)
- $\eta = 5$ is set such that the individual labor supply elasticity is in the range of the values reported by MaCurdy (1981).
- The value of γ allows to replicate the 15% stationary ratio of imports to GDP.

3. Calibration

C. Labor market

Mainly follow Andolfatto (1996)

- The values of σ_n and σ_u are chosen in order to get stationary values of 0.57 for N , the employment ratio, and of $1/3$ for h , the average fraction of time spent working.
- Expenditures on search activity are fixed at 1% of GDP ($\omega_i V_i / Y_i$)
- Average fraction of time that nonemployed households devote to search is set to $1/2h$.
- χ is fixed to get a value of 0.9 for the probability that a vacant position becomes a productive job (see the matching function).
- ϕ is at the value 0.60, estimated by Blanchard and Diamond (1989). We set ϵ at the same value, implying that Hosios's (1990) condition is satisfied: following Merz (1995) and Andolfatto (1996), wage and hours bargaining is such that trade externalities do not distort equilibrium.
- The rate of transition from employment to nonemployment s is set at 0.10, according to Davis et al. (1996).

4. Simulation

A. Moments

Cross-country Business cycle (with US)

	Output	Labor	Investment	Consumption
Data	0.51	0.36	0.38	0.40
Search labor market	0.29	0.25	0.08	0.71
Walrasian labor market	0.25	-0.18	-0.19	0.80

- Outputs of developed countries appear to move together.
- The correlation is larger for output than for consumption, the latter being roughly similar to that of investment and labor.
- The inability of traditional international RBC models to reproduce these features has been much documented, especially the international input co movements, when only technology shocks are taken into account.

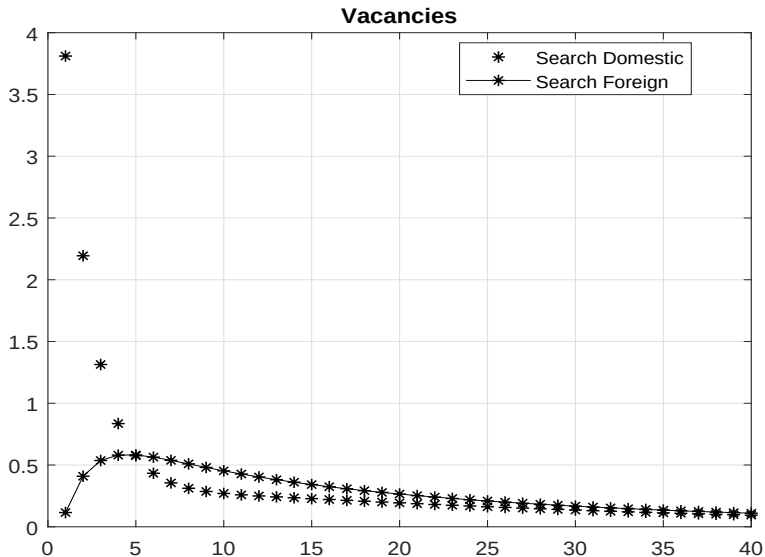
4. Simulation

A. Moments

- The two-country search model appears more in accordance with the empirical correlations, even if it does not perfectly succeed in replicating the observed magnitude of international co movements. Introducing search leads to a positive co movement for total hours and a non-negative one for investment.
- These results derive from the particular employment behavior in the search model, whereby the dynamics are indeed roughly similar for employment and total hours.

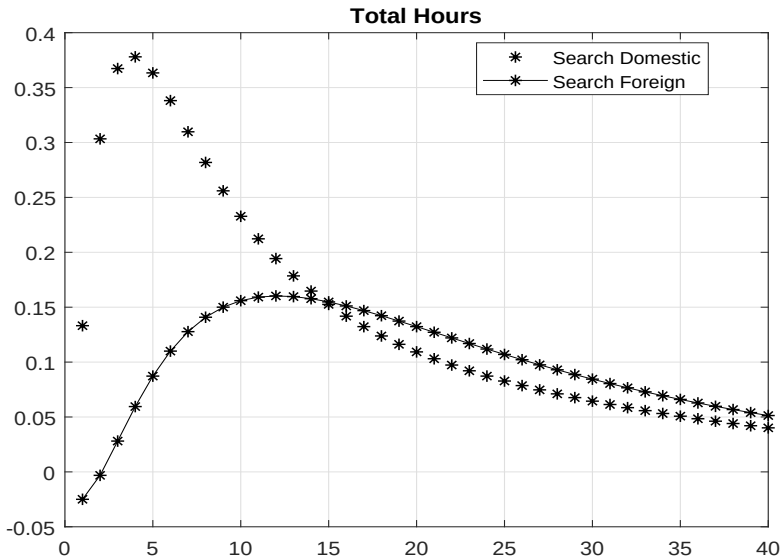
4. Simulation

B. Impulse response function



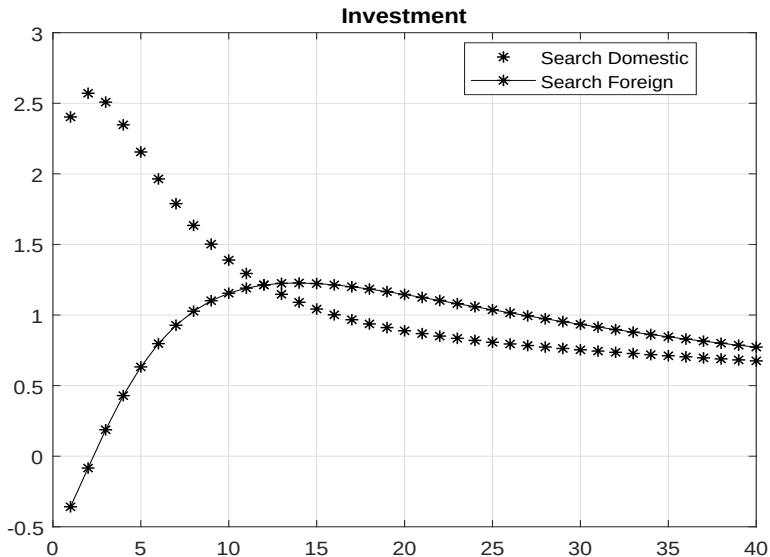
4. Simulation

B. Impulse response function



4. Simulation

B. Impulse response function



4. Simulation

B. Impulse response function

- The simulation results can be explained by computing the impulse response functions following a productivity shock in country 1.
- The key mechanism for understanding positive aggregate co movements refers to the search process on the labor market.
- The technology shock spills over to the other country, domestic and foreign firms start searching at the same time in anticipation
- Total hours rise slowly in the two countries , displaying a typical hump-shaped response.
- Dynamic totally different in a Walrasian model (Search frictions)
- The capital investment, which exhibits a hump-shaped dynamics improve the replication of the observed investment correlations across countries.
- Higher employment partially curtails the capital outflow from the second country by increasing the capital productivity. Foreign capital remains almost unchanged in country 2 during the first periods and increases afterwards.

4. Extensions

C. Non separable utility function

$$U_i^s = \log(C_i^{c,s} + \Gamma_s(h_i^s)), \quad s = n, u \quad i = 1, 2$$